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25 February 1980
SUPERSEDING
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12 September 1961
(See 6.6 and 6.7)

MILITARY SPECIFICATION

DECK COVERING COMPOUND, NONSLIP, ROLLABLE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rollable, nonslip, lightweight deck coverings for application to metal decks.

1.2 Classification. The total deck covering system consists of a primer and a nonslip topcoat (base resin, nonslip aggregate, and hardener). Topcoats shall be furnished in the following types, as specified (see 6.2.1):

Type III - General purpose exterior.
Type IV - Aircraft carrier landing and run-out area (nonabrasive to steel arresting cable).
Type V - General purpose interior or exterior (high flash point).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-E-760 - Ethyl Alcohol (Ethanol); Denatured Alcohol; Proprietary Solvents and Special Industrial Solvents.
PP-C-96 - Can, Metal, 28 Gage and Lighter.
QQ-W-470 - Wire, Steel, Carbon, Spring, Music.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).

MILITARY

MIL-H-5606 - Hydraulic, Fluid, Petroleum Base; Aircraft, Missile and Ordnance.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-D-16791 - Detergents, General Purpose (Liquid), Nonionic.
MIL-G-18709 - Grease, Ball and Roller Bearing.
MIL-P-24441 - Paint, Epoxy-Polyamide, General Specification for.
MIL-P-24441/1 - Paint, Epoxy-Polyamide, Green Primer, Formula 150.

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials; Methods for Testing of.
FED-STD-313 - Material Safety Data Sheets, Preparation and Submission of.
FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.
MIL-STD-1623 - Fire Performance Requirements and Approved Specifications for Interior Finish Materials and Furnishings (Naval Shipboard Use).

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 93 - Flash Point by Pensky-Martens Closed Tester, Test for.
D 1141 - Substitute Ocean Water.
D 1652 - Epoxy Content of Epoxy Resins, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA Traffic Dept., 1616 "P" Street, N.W., Washington, DC 20036.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT
Rules and Regulations - Rule 102.

(Application for copies should be addressed to the Los Angeles County Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

(Technical society and technical association specifications and standards are generally available or reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Deck covering compound furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.5).

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3.2 Description. The nonslip covering shall be applied with a smooth, hard core roller as specified by the manufacturer. After the nonslip has been applied and cured, the surface shall have a textured appearance of roughly parallel rows of raised coating material forming peaks or ridges with the aggregate presenting a uniformly, coarse, rough appearance over the entire surface. The overall surface appearance shall present a wave-like ridged pattern with raised peaks. The nonslip shall have sufficient "sag resistance" to retain its raised, textured appearance after application.

3.3 Composition. The deck covering "system" shall consist of a primer and a topcoat, the latter to include a nonslip aggregate. The primer shall conform to MIL-P-24441/1, formula 150. The topcoat shall be a two part synthetic resin compound consisting of the base resin plus aggregate, and hardener (components A and B respectively, see 5.1.1).

3.3.1 The deck covering compound shall be free of any asbestos component. A certificate of compliance shall be required that the deck covering compound is asbestos-free (see 6.2.2).

3.4 Condition-in-container. When mixed in accordance with manufacturer's instructions, the components of the topcoat shall be readily dispersed to a uniform consistency, shall not liver, gel, or show any other objectionable properties which will adversely affect performance, and shall produce a nonslip covering which meets all the requirements of this specification for at least one year after the date of delivery (see 4.6.19).

3.5 Nonvolatile content. When tested as specified in 4.6.2, the minimum nonvolatile content of the topcoat deck covering shall be 70 percent.

3.6 Pot life. When tested as specified in 4.6.3, the minimum pot life shall be 4 hours minimum, for types III and IV; 1.5 hours minimum for type V.

3.7 Thickness. When tested as specified in 4.6.4, the average overall minimum thickness, including aggregate, when cured, shall be 30 mils.

3.8 Weight. When tested as specified in 4.6.4, the weight shall be not less than 4.5 nor more than 6.5 ounces per square foot.

3.9 Coverage. When applied as specified in 4.6.5, the minimum coverage shall be 30 square feet per gallon, minimum, and 50 square feet per gallon, maximum.

3.10 Fire resistance. The deck covering shall conform to the fire test requirement specified in MIL-STD-1623 (see 4.6.12).

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3.11 Resistance to impact. When tested as specified in 4.6.6, the deck covering shall have a minimum rating for all tests of 95 percent except for impact after seawater immersion where the minimum rating shall be 90 percent.

3.12 Abrasion of arresting gear cable (type IV only). When tested as specified in 4.6.7, the type IV covering shall produce a maximum average wear of the test wire of 0.4 mils.

3.13 Nonslip properties. When tested as specified in 4.6.8, the coefficients of static friction for the deck covering shall not be less than 0.90 for dry, wet, or oily conditions. For type IV only, the coefficient of friction (COF) shall be conducted after the cable wear test specified in 4.6.7. The COF shall not be less than 1.00 dry; 0.95 wet or oily.

3.14 Resistance to wear. When tested as specified in 4.6.9, the deck covering shall produce a maximum reduction in thickness of 30 mils. In worn areas, the evenly distributed aggregate should continue to protrude above the resin matrix.

3.15 Resistance to sea water immersion and temperature changes. When tested as specified in 4.6.10, the deck covering system shall show no signs of cracking, corrosion of the steel substrate, or separation of the primer-topcoat interface.

3.16 Resistance to immersion. When tested as specified in 4.6.11, the deck covering system shall show no signs of softening, loss of adhesion at the steel-primer and primer-topcoat interfaces, discoloration, or any other form of deterioration.

3.17 Flash point. When tested as specified in 4.6.13, the mixed ingredients for types III and IV shall have a minimum flash point of 80° Fahrenheit (F); for type V, the mimimum flash point shall be 300°F.

3.18 Drying time. When the components of the topcoat are mixed and applied as specified in 4.6.1, the topcoat shall have a maximum drying time of 24 hours.

3.19 Color. Topcoats shall be available in the following colors: yellow to match No. 33538; red to match No. 31136; olive drab to match No. 24087; haze gray to match No. 36270; dark gray to match No. 36076; black to match No. 37038, and white to match No. 37875 of FED-STD-595 (see 4.6.14, 6.2.1, and 6.4).

3.20 Identification characteristics. Values for identification characteristics shall be provided by individual contractors for characteristics as indicated by "X" in table I.

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TABLE I. Identification characteristics values.

Characteristic	Base material, component A	Curing agent, component B	Combined mixture	Aggregate
Chemical nature	X	X	---	X
Percent of each principal constituent (5 percent or more of total) by weight	X	X	---	X
Percent pigment plus aggregate (if applicable), by weight	X	X	---	---
Percent volatile, by weight	X	X	---	---
Percent nonvolatile vehicle, by weight	X	X	---	---
Weight per gallon	X	X	X	---
Consistency	X	X	X	---
Flash point	X	X	X	---
Pot life:				
at 90°F	---	---	X	---
at 50°F	---	---	X	---
Coverage, square feet per gallon (ft^2/gal)	---	---	X	---
Drying or curing time, hours	---	---	X	---
Mixing instructions	---	---	X	---
Application instructions	---	---	X	---
Aggregate sizes (sieve test)	---	---	---	X
Epoxy content (ASTM D 1652)1/	X	---	---	---
Compliance with air pollution requirements2/	Certify3/ Yes or No			
Asbestos4/	X	X	---	X

1/ Polyurethane content, if applicable.
2/ If yes, provide data on solvent composition.
3/ Formulation must comply with the requirements of Rule 102 of the South Coast Air Quality Management District, Los Angeles, CA.
4/ Product must be certified to be free of asbestos.

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3.21 Directions for mixing and applying. Written directions for mixing, rolling or trowelling as recommended by the manufacturer, including working life, application temperature limitations, and safety precautions, shall be furnished with each contract, as specified in 6.2.1. Directions which differ from those furnished under 3.20 shall be submitted to the Naval Sea Systems Command for review.

3.22 Serviceability. The deck covering systems shall show no loss of adhesion or other deficiency which would limit their serviceability when examined after a minimum of six months for types III, IV, or V materials, or after a minimum of 1500 landings for type IV materials (see 4.6.17).

3.23 Service use. The types III and V deck covering systems shall continue to provide satisfactory performance without excessive touch-up for a minimum period of one year (see 4.6.17).

3.24 Resistance to accelerated light and weather aging. When tested as specified in 4.6.18, the exposed surface on the covering shall show no appreciable change in color, signs of checking, cracking, or any other deterioration.

3.25 The nonslip coating shall conform to all the requirements specified in table II.

TABLE II. Requirements.

Item	Test requirements			Test method paragraph	Applicable paragraph
	Type III	Type IV	Type V		
Drying time, hours, maximum	24	24	24	4.6.1	3.18
Condition in container, years, minimum	1	1	1	4.6.19	3.4
Nonvolatile content, percent, minimum	70	70	70	4.6.2	3.5
Pot life, hours, minimum	4	4	1.5	4.6.3	3.6
Thickness, mils, minimum	30	30	30	4.6.4	3.7
Weight, ounces per square foot:					
Minimum	4.5	4.5	4.5	4.6.4	3.8
Maximum	6.5	6.5	6.5		

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TABLE II. Requirements. - Continued

Item	Test requirements			Test method paragraph	Applicable paragraph
	Type III	Type IV	Type V		
Coverage, ft^2/gal:					
Minimum	30	30	30		
Maximum	50	50	50	4.6.5	3.9
Impact resistance:					
Initial and after aging percent, minimum	95	95	95	4.6.6	3.11
After seawater immersion, percent, minimum	90	90	90	4.6.10	3.15
Abrasion resistance of cable, mils, maximum (type IV only)	----	0.4	----	4.6.7	3.12
Nonslip properties:					
Dry, wet, or oily	0.90	0.90	0.90	4.6.8	3.13
Nonslip properties, after cable wear test (type III only):					
Dry	----	1.00	----	4.6.7 and	3.13
Wet and oily	----	0.95	----	4.6.8	
Wear resistance, mils, maximum	30	30	30	4.6.9	3.14
Moisture and temperature resistance	Shall pass test			4.6.10	3.15
Immersion resistance	Shall past test			4.6.11	3.16
Fire resistance:					
Average char, inches, maximum	See MIL-STD-1623			4.6.12	3.10
After-flaming	----	None	----		
Flash point, degrees F, minimum	80	80	300	4.6.13	3.17
Color	To match paint chip			4.6.14	3.19
ID characteristics	(See table I)			------	3.20
Serviceability	Shall pass test			4.6.17	3.22
Service use	Shall pass test			4.6.17	3.23
Accelerated light and weather aging	Shall pass test			4.6.18	3.24

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3.26 Material Safety Data Sheet. The contracting activity shall be provided a material safety data sheet (MSDS at the time of contract award. The MSDS is DD Form 1813 and found as part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Verification tests (see 4.4.4).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command. Qualification tests shall consist of the examination of 4.4.2 and the tests specified in 4.6. Service test shall be conducted on a carrier flight deck in accordance with 4.6.17.

4.3.1 Qualification samples. For qualification, the contractor shall submit to the designated laboratory, one pint container of the nonslip aggregate ingredient, one gallon container of the base material, component A (with aggregate as applicable), and one gallon of curing agent, component B, each packaged in a separate container.

4.3.2 Qualification approval for type IV shall also consititute approval for type III; however, approval for type III does not constitute approval for type IV.

4.4 Quality conformance inspection.

4.4.1 Lot. For purposes of examination and testing, a lot shall consist of a production batch of topcoat that is offered for delivery under a contract or order.

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4.4.1.1 Sampling for quality conformance inspection.

4.4.1.1.1 Sampling for examination of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) of 2.5 percent defective for the examination specified in 4.4.2.

4.4.1.1.2 Sampling for quality conformance tests. From each lot of material for delivery, a two-quart sample of base material with sufficient curing agent, shall be selected at random for the quality conformance tests specified in 4.4. Where lots are small, only one 2-quart sample of base material shall be selected for 500 gallons of material.

4.4.1.1.3 Sampling for verification tests. From the first lot of a single type offered for delivery under a contract or order, and for every 5,000 gallons thereafter, one 2-quart sample of base material with sufficient curing agent, shall be selected at random and forwarded to a designated Government laboratory for verification tests.

4.4.2 Examination. Each filled container in the sample selected in accordance with 4.4.1.1.1 shall be examined to verify compliance with this specification. Any container in the sample having one or more defects under required fill shall not be offered for delivery. If the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105 (see 4.4.1.1.1), this shall be cause for rejection of the lot represented by the sample.

4.4.3 Quality conformance tests. Quality conformance tests for acceptance of individual lots shall consist of all tests specified in 4.5.1.

4.4.4 Verification tests. Verification tests shall consist of any tests determined to be necessary by the Naval Sea Systems Command for conformance to this specification.

4.5 Acceptance of individual lots. Acceptance of individual lots shall be based on compliance of material with the desired results obtained by 4.5.1.

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4.5.1 Lot acceptance tests. The following tests shall be performed and results shall meet the applicable requirements in section 3.

	Test	Requirement
Drying time	4.6.1	3.18
Nonvolatile content	4.6.2	3.5
Pot life	4.6.3	3.6
Thickness	4.6.4	3.7
Weight	4.6.4	3.8
Resistance to impact	4.6.6.1(a), (b), and (c)	3.11
Nonslip properties (initial)	4.6.8	3.13
Flash point	4.6.13	3.17
Color	4.6.14	3.19

4.6 Test procedures.

4.6.1 Preparation of specimens. Specimens of the size specified in the following tests shall be prepared, using 1/8-inch thick mild steel plates except for the 1/16-inch thick mild steel plates used for thickness and weight as specified in 4.6.4, and for nonslip and wear as specified in 4.6.8 and 4.6.9, respectively. Panels shall be blasted to a uniform white appearance with a minimum surface profile of 1.0 - 2.0 mils peak to valley. Panels shall be cleaned after blasting, using clean, dry compressed air, vacuum, or solvent mixture of 1:1 xylene and isopropanol and primed 2 to 4 mils dry film thickness with F-150 of MIL-P-24441/1. The nonslip coating shall be mixed and rolled in one coat in accordance with the procedure recommended by the manufacturer. The coated steel specimens shall cure for a minimum of 96 hours before conducting tests. Unless otherwise specified herein, all curing and testing shall be conducted under atmospheric conditions at a temperature of 72°F ± 3°F, and a relative humidity of 50 ± 2 percent.

4.6.1.1 Drying time. The dry-hard time shall be determined by method 4061.1 (dry-hard) of FED-STD-141.

4.6.2 Nonvolatile content. The nonvolatile content of the mixed product (resin and hardener) shall be determined in accordance with method 4042 of FED-STD-141, except that the sample weight shall be doubled and the 5 milliliter (mL) syringe replaced with a more practical dispensing method.

4.6.3 Pot life. Mixing procedures shall be in strict accordance with the manufacturer's instructions. The packaged ingredients comprising the components of the topcoat shall be thoroughly mixed by hand and allowed to stand in an open

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container at a temperature of 75°F ± 3°F and a relative humidity of 50 ± 2 percent. The components shall be placed in a quart container in sufficient quantity to fill the container to approximately 1/2 inch from the top. The material shall be rollable within the specified pot life period.

4.6.4 Thickness and weight. A set of three mild steel plates, 6 by 6 inches, 1/16 inch thick shall be prepared as specified in 4.6.1. The thickness and weight of each shall be measured. The weight shall be determined to the nearest 0.1 gram. The coating thickness shall be determined at 16 equally distributed points on each of the plates by means of a dial type thickness gage and template. The nonslip material shall be rolled one coat in accordance with the manufacturer's instructions so as to completely cover the steel plates, and allowed to cure for a minimum of 96 hours. After curing the three test specimens, each shall be weighed to the nearest 0.1 gram. The length and width shall be measured to the nearest 0.01 inch and the thickness to the nearest 0.001 inch at each of the points previously measured. The difference in thickness of the uncoated steel plate and the coated steel plate at each of the points shall be determined and the results averaged for each of the specimens. The thickness of each specimen shall then be averaged and this value reported as the coating thickness. The weight of the coating shall be determined from the difference between the weight of the coated steel plate and the uncoated steel plate. The weight of the three specimens shall be averaged and multiplied by four to convert to grams per square foot (to convert to ounces per square foot, multiply grams per square foot by 0.035).

4.6.5 Coverage. Coverage shall be determined by applying the topcoat to a 3 foot by 3 foot by 1/8-inch thick or 1/16 inch thick (optional) primed mild bare steel plate in accordance with the application procedure recommeded by the manufacturer to produce mixing and a minimum thickness of 30 mils (see 4.6.4). Result shall be the average of at least two determinations.

4.6.6 Resistance to impact.

4.6.6.1 Procedure. A set of three specimens, 6 inches by 6 inches, shall be prepared as specified in 4.6.1 to represent each of the treatments below. Topcoat shall be applied over the primer within 24 hours after it is "dry to touch".

(a) No treatment, or "as is"
(b) 4 hours at 32°F
(c) 4 hours at 158°F
(d) 15 day immersion in synthetic sea water

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4.6.6.2 Immediately upon removal from treatment, the specimens shall be subjected to 25 impacts by a 2-pound solid steel ball dropped from a height of 8 feet. The steel ball shall be held in a fixed position by an electromagnet provided with a centering tip. The specimen shall be held on a 1-1/2-inch thick steel base which shall be fitted with guides to reposition the specimen after each impact, so that successive points of impact are equally spaced 3/4 inch apart for 25 impacts in a 5 by 5 pattern (5 rows of 5 impacts per row) enclosed within an area of approximately 9 square inches.

4.6.6.3 Rating. Upon completion of each impact test, the test specimen shall be examined for failure of the covering between impacts. The evaluation will be based on a percent estimate of the portion of the covering between impact sites that remain intact. A rating of 100 percent would show 25 discrete impacts with the covering around each impact firm and adhering to the substrate. A rating of 0 percent would show all the covering removed within the 9 inch square area.

4.6.7 Arresting gear cable wear (type IV only). Three specimens 8 inches long by 4 inches wide shall be prepared as specified in 4.6.1. Each specimen shall be subjected to 200 cycles of wear by an 1/8-inch diameter music wire conforming to QQ-W-470. Each specimen shall be evaluated with a new wire. The diameter of each wire shall be measured. The wire shall then be held in a jig which is clamped in the test apparatus (see 4.6.7.1) shown on figure 1. After completion of 200 cycles in the wear test machine, the wire shall be removed from the jig and the amount of wear on the wire shall be determined by measuring the thickness of the wire. Measurements shall be made to the nearest 0.0001 inch by means of a dial type thickness gage having a foot fitted with a knife edge which shall contact the worn surface of the wire, and a 1-inch diameter flat anvil on which the wire shall be supported. The average of ten measurements for each three wires shall be taken as the average wear caused by the topcoat. The above three specimens shall be saved and later used to determine the nonslip properties for the "after cable" wear test (see 4.6.8.1).

4.6.7.1 Arresting gear cable wear test apparatus. The test apparatus as shown on figure 1 shall have essential characteristics as follows:

(a) A reciprocating carriage capable of a throw of 6 inches per cycle, at a frequency of approximately 30 cycles per minute and shall be equipped with clamps to hold the 8-inch by 4-inch specimen.

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(b) A pivoted clamp assembly arranged to support a jig for holding an 8-inch length of 1/8-inch diameter music wire with its axis horizontal and at right angle to the direction of motion of the reciprocating carriage. The jig shall consist of an 8-inch long, 1-inch diameter steel bar into which a 1/6-inch "V" notch has been cut along its length for holding the wire in contact with the nonslip deck coating without flexing or twisting. The pivot shall be 12-1/2-inches from the wire held in the "V" notch of the jig, and the entire pivoted assembly (including the test wire and jig) shall be weighted such that the wire shall contact the coated specimen under a load of 20-1/4 pounds. The test wire in the jig shall be clamped at each end so that no part of the coated clamp contacts the coated specimen.

4.6.8 Nonslip properties (coefficient of static friction). Thirty-six numbered, 2-inch by 3-inch by 1/16-inch specimens shall be prepared as specified in 4.6.1, butted to fashion a 6 by 6 array on a plywood board, forming a 12-inch by 18-inch test surface. After drying and curing, the specimens are separated and excess coating scraped off their edges. Randomly select nine specimens for evaluation of static friction over dry, synthetic seawater wet, and oily surfaces. The specimens shall be evaluated for static friction after 350 evolutions in the wear test machine in the following order and surface conditions, dry, wet, and oily. For static friction over wet surfaces, the specimens are wet with synthetic seawater prepared from sea salt in accordance with ASTM D 1141. The specimens shall be rinsed in tap water to remove the synthetic seawater, blotted with paper towels, dried in an oven at 250°F for 1 hour, and then permitted to cool to ambient temperature prior to testing at the oily surface condition obtained by wetting with Society of Automotive Engineers (SAE) 10W oil at a temperature between 60°F and 80°F.

4.6.8.1 Procedure. The coefficient of static friction shall be determined by means of a properly calibrated Olson Slipmeter, as shown on figure 2. The slipmeter is a portable instrument with a 6-pound, weighted meter that is supported by a set of three removable, 3/4-inch diameter feet. Two sets of feet shall be fitted with 1/8-inch thick vulcanized neoprene rubber having a Shore "A" hardness range of 85 ± 5 durometers. Precondition the rubber surfaces with sandpaper prior to the static friction test by rubbing a piece of 80 grit sandpaper in at least two directions at right angles to expose fresh rubber. Obtain "afterwear" data after 350 revolutions in the wear test machine by pulling the slipmeter over three specimens at a rate of 4.0 inches per minute by means of a cord wound over

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a constant speed motor-driven winch in front of the specimens. A set of three 2-inch by 3-inch specimens shall be held along their 3-inch axis and arranged as shown on figure 3, such that each foot of the slipmeter contacts a separate specimen. For each of the three readings, the three specimens support a different foot of the slipmeter. The specimens are then rotated 180 degrees and the procedure repeated for a total of six readings. Static friction measurements are repeated for the second and third sets. A total of 18 readings are made for each surface condition. Each of the 4-inch by 8-inch specimens (see 4.6.7) shall be held lengthwise so that all three feet of the slipmeter are in contact with the specimen during tests. One reading shall be recorded on each of the three 4-inch by 8-inch specimens after which the specimens shall be rotated 180 degrees and readings for each specimen shall be again recorded for a total of six readings. "After-wear" by the wear test machine (2-inch by 3-inch specimens) (see 4.6.9), and by the arresting gear cable wear test apparatus (4-inch by 8-inch specimen) (see 4.6.7), the static friction shall be determined with the slipmeter for each of three surface conditions, dry, wet with synthetic seawater, and oily. One set of rubber feet shall be used for the dry and wet surface conditions and another set of rubber feet shall be used for the oily surface condition.

4.6.9 Resistance to wear. Nine specimens, 2-inches by 3-inches, prepared as specified in 4.6.1, shall be mounted so that an area of approximately 6 square inches of the coating systems is presented to the abrasive on the disk of the wear test machine described herein. The abrasive shall be number 80 aluminum oxide. The thickness shall be measured with a dial type thickness gauge at 12 equally distributed points positioned by use of a template. The specimens are subjected to an initial 50 revolutions in the wear test machine to remove asperities prior to taking thickness measurements. The specimens shall then be subjected to an additional 300 revolutions of the wear-testing machine. The difference in thickness shall be averaged to determine the thickness of the coating material worn away by the abrasive.

4.6.9.1 Wear-test machine. The wear-test machine as shown on figures 4 and 5 shall have the following essential mechanical characteristics:

(a) Mild steel, abrasion disk 14-1/8-inches in diameter and 1/2-inch thick, revolved clockwise at a constant speed of 23.5 revolutions per minute (r/min).
(b) Specimen holder, attached to a shaft. The distance between center of the disk and the specimen holder shaft shall be 5.1 inches. The holder shall revolve clockwise at a constant speed of 32.5 r/min. In each one-half revolution

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of the shaft, the specimen shall be lifted a distance of 1/16-inch and then dropped back on disk (a) under a 10-pound weight. Change in the lift due to wear of the specimen shall be compensated by adjustment of a cam at regular intervals, so that the lift is maintained at a practically constant distance.

(c) Weight, 10 pounds is the maximum found suitable for use with this test machine.

(d) Distribution of abrasive grit. The grit flows by gravity from a hopper through a 13/16-inch inside diameter tube and into the center of a distribution chamber. An impeller in the chamber provides an evenly distributed feed of the grit through eight 3/64-inch diameter holes center on a 1-5/8-inch diameter pitch circle. A uniform distribution of grit shall fall to disk (a) in the path of the test specimen. The distributor shall be revolved at a constant speed of 9 r/min. Details of the distributor for the abrasive grit, the cam and specimen holder are shown on figures 6 and 7, respectively.

4.6.9.2 Procedure. The steel disk (a) shall revolve while the weighted specimen of the material in holder (b) shall be lifted and dropped by a cam (c) while it is being revolved. When the specimen is dropped through motion of the cam, it shall fall a distance of 1/16-inch and meet disk (a) with impact simulating the fall of a foot on the deck surface. Foot action shall be further simulated in that the revolving specimen shall contact the disk with a sliding and twisting motion until lifted clear by the cam. Continuous rotation of the specimen shall introduce the element of wear in all directions on the material. Free abrasive grit of number 80 aluminum oxide shall be fed at a steady rate from the hopper and distributed uniformly on the steel disk in the path of the specimen. The revolving motions of the steel disk and the test specimen shall keep the abrasive grit circulating to the outer side of the path of the specimen. The grit shall fall from the edge of the steel disk into a hopper from which it is collected and then screened to eliminate undesired particles before replacement in the feeding hopper.

4.6.10 Resistance to seawater immersion and temperature changes. Tests shall be made on three 2-inch by 6-inch specimens prepared as specified in 4.6.1. Each specimen shall be immersed for 48 hours in synthetic seawater prepared from sea salt in accordance with ASTM D 1141, under a pressure of 8 pounds per square inch (lb/in^2). Immediately after removal from immersion, each specimen shall be subjected to two complete cycles of alternate exposure to a temperature of $0^{\circ}F \pm 5^{\circ}F$, for 24 hours, followed by a temperature of $120^{\circ}F \pm 5^{\circ}F$ for 24 hours. The

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specimen shall then be examined for evidence of cracking or other failure. A portion of the coating shall be carefully removed from the steel plate to observe any signs of rusting or corrosion under the coating.

4.6.11 Resistance to immersion. Eighteen 2-inch by 6-inch specimens shall be prepared as specified in 4.6.1. Three specimens shall be positioned vertically with respect to their 6-inch length and exposed for 24 hours to each one of the six media listed below. Only the bottom half of the specimen shall be immersed in the media.

Test media

(a) SAE 10W oil.
(b) Grease conforming to MIL-G-18709.
(c) Jet fuel JP-5 conforming to MIL-T-5624.
(d) Hydraulic fluid conforming to MIL-H-5606.
(e) Ethyl alcohol conforming to O-E-760, type III.
(f) Detergent conforming to MIL-D-16791, 1/2 percent solution.

4.6.11.1 Upon removal from the immersion medium, the test specimens shall be examined for softening, loss of adhesion at the primer-topcoat interfaces, and other forms of deterioration.

4.6.12 Fire resistance test. The material shall be tested in accordance with the test procedure set forth in MIL-STD-1623.

4.6.13 Flash point. The flash point shall be determined immediately after mixing in accordance with ASTM D 93.

4.6.14 Color. A specimen 6 inches by 6 inches, prepared as specified in 4.6.1, shall be compared with the standard color chip (see 3.19, 6.2.1, and 6.4).

4.6.15 Aggregate. Aggregate shall be equivalent to the samples of the original qualified aggregate material visually; and, if required, by chemical or physical analysis.

4.6.16 Packaging inspection. Sample packages and packs, and the inspection of the packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.6.17 Serviceability and service use. A service test shall be conducted on a carrier flight deck (landing area or nonlanding area, as applicable). Material shall be roller applied at the coverage and thickness in accordance with the

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manufacturer's mixing and application instructions. For types III and V materials, performance shall be determined at 6 months and 12 months to determine compliance with 3.22 and 3.23. For type IV material, performance shall be determined after 1500 landings.

4.6.18 Accelerated light and weather aging. Test specimens, 6 inches by 2 inches, prepared as specified in 4.6.1, shall be exposed in an ATLAS mode XW or equivalent unit for 200 hours, and intermittently sprayed with tap water. The face of the material normally exposed in service shall be directed toward the light source. The exposed surface of the test specimen shall be examined and compared with an unexposed specimen of the same size with respect to discoloration, cracking, checking, crazing, or other signs of deterioration.

4.6.19 Condition in container. All components of deck covering compound after being sealed in their original containers and held in storage for a minimum period of one year, between 50°F and 90°F, shall conform to 3.4, and the tests specified in 4.5.1.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions.)

5.1 Packaging.

5.1.1 Levels A and C. The topcoat shall be supplied as a kit, consisting of base material marked component A, and curing agent marked component B. The quantity of each component in the kit shall be in the proper proportions when mixed together to meet the performance requirements herein. Component A shall be furnished in 1-gallon cans or pails or in 5-gallon pails as specified (see 6.2.1). Component B shall be furnished in cans or pails of appropriate size to contain the curing quantity for over 5 gallons of base resin. Alternatively, the topcoat kit, components A and B, may be packaged in a 6- or 7-gallon pail provided with an auxiliary compartment.

5.1.1.1 Cans. Cans shall conform to type V, class 2 of PPP-C-96. Interior of cans may be unlined; plan B coating and side seam stripping are required. Wire handles treated to resist corrosion are required for 1-gallon cans.

5.1.1.2 One- and five-gallon pails. One- and five-gallon pails shall conform to type II, class 3 of PPP-P-704. Interior coating is not required. Wire handles on pails shall be treated to resist corrosion.

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5.1.1.3 Double-compartment pail.

5.1.1.3.1 Six-gallon pail. The 6-gallon pail shall conform to type II, class 5 or 11 at the contractor's option of PPP-P-704. Interior coating is not required. Wire handles or bails shall be treated to resist corrosion.

5.1.1.3.2 Auxiliary compartment. The auxiliary compartment may be a tray or of saddle configuration; fabricated of metal or plastic and designed to rest over the pail edge. Compartments shall be provided with a gasket(s) to provide a snug fit and leak-proof closure when pail lid is secured. The compartment shall be of sufficient size to accommodate the packaged quantity of curing agent, component B.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Levels A and B.

5.2.1.1 Cans. Each kit, packaged as specified, shall be packed in a fiberboard box conforming to PPP-B-636, class weather resistant for level A and class domestic for level B. Fiberboard pads shall be placed on the bottom and tops of the cans. For level A, boxes shall be closed and waterproofed in accordance with method V; for level B, boxes shall be closed in accordance with method I as specified in the appendix to the box specification. Reinforcing of level A boxes shall be accomplished by the use of nonmetallic banding or pressure sensitive reinforced tape at the contractor's option.

5.2.1.2 Pails. One-gallon pails shall be packed for level A and B as specified in 5.2.1.1. Pails require no further packing for levels A and B. When specified (see 6.2.1), pails shall be palletized in accordance with MIL-STD-147.

5.2.2 Level C. Deck covering, packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Container packing or method of shipment shall comply with Uniform Freight Classification Ratings, Rules, and Regulations or National Motor Freight Classification or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2.1) or herein, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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5.3.1 Special marking. Each component container shall have the following additional information:

(a) Type.
(b) Manufacturer's name.
(c) Manufacturer's lot number and date of manufacture.
(d) Contract number.
(e) Color designation and number.
(f) Shelf-life marking.
(g) For products indicating compliance with the air pollution requirement in table II, the following additional marking is required:

"The volatile content of the material in this container is not photochemically reactive as defined by Rule 102 of the South Coast Air Quality Management District."

(h) Labeling. The major bold lettering on labels attached to the pails shall be red on a white background for type III; blue on a white background for type IV; green on a white background for type V.

5.3.1.1 Mixing, rolling, trowelling, and use instructions. Each component container shall be marked with the following:

"CAUTION

This is one component of a two component system which WILL NOT HARDEN unless both components are mixed together.

INSTRUCTIONS FOR USE

Refer to manufacturer's written mixing and application instructions prior to use."

5.3.1.1.1 Additional types III and V marking. For types III and V kits, containers shall be marked on lid and container with the following in addition to other requirements.

"DO NOT USE IN CARRIER LANDING AREA"

5.3.1.1.2 Air pollution compliance. For those products which have been approved for use in areas with regulations controlling the emission of solvents into the atmosphere (see 3.20 and 6.3), containers shall include the following statement: "Complies with air pollution regulations."

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5.3.1.2 Caution label. Each component container shall be marked with the following:

"CAUTION: Avoid skin contact during application. Chemicals may cause irritation or skin sensitization. In case of contact, wash skin thoroughly with soap and water."

5.3.1.3 For shipping containers and palletized unit loads, a copy of the Material Safety Data Sheet (DD 1813) shall be attached to the shipping document for each destination (see 3.26). The contractor shall furnish a copy of the DD 1813 to the Naval Sea Systems Command with application for product qualification.

5.4 Primer. The primer shall be prepared for shipment in accordance with the requirements of MIL-P-24441.

6. NOTES

6.1 Intended use. These materials, when rolled over a primed metal (or wood) deck, are intended to provide slip resistance for aircraft, vehicles and personnel under dry, oily, or wet conditions.

6.1.1 Type III is a general purpose type. It is intended for use aboard ship where a slip resistant surface is needed such as flight and hangar decks on carriers, except beneath the arresting cables and runout area.

6.1.2 Type IV is specially formulated as a nonabrasive to be used beneath the steel arresting cables on carrier flight decks. It may be used elsewhere aboard ship where a slip resistant surface is required.

6.1.3 Type V is a high flash point slip resistant material designed for interior or exterior use where a low fire risk material (during application) is required.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) Type of deck covering (see 1.2).
(c) Color of material (see 3.19 and 4.6.14).
(d) Compliance with air pollution regulations required (see table I).

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(e) Number of copies of manufacturer's written directions for mixing and applying (see 3.21).
(f) Quantity of kits and size of component containers (see 5.1).
(g) Type container for 1-gallon quantity: can or pail (see 5.1).
(h) Level of packaging and packing required (see 5.1 and 5.2).
(i) Palletization, when required (see 5.2.1.2).
(j) Special markings required (see 5.3).

6.2.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
3.3.1	Certificate of Compliance	DI-E-2121	------

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Formulations which comply with air pollution regulations may be required for use in areas with regulations controlling the emission of solvents into the atmosphere.

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6.4 Color chips may be obtained from General Services Administration (GSA), Specification Sales, (Code 3FRI), Building 197, Washington Navy Yard, Washington, DC 20407.

6.5 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List QPL 23003 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5.1).

6.5.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.6 Types I and II of MIL-D-23003(SHIPS) are obsolete. The deck covering system consisting of a primer and a nonslip topcoat (base resin, nonslip aggregate, and hardener) shall be furnished in the following types (see 1.2):

Type III - General purpose exterior
Type IV - Aircraft carrier landing and run-out area (nonabrasive to steel arresting cable)
Type V - General purpose interior or exterior (high flash point)

6.7 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5610-N053)

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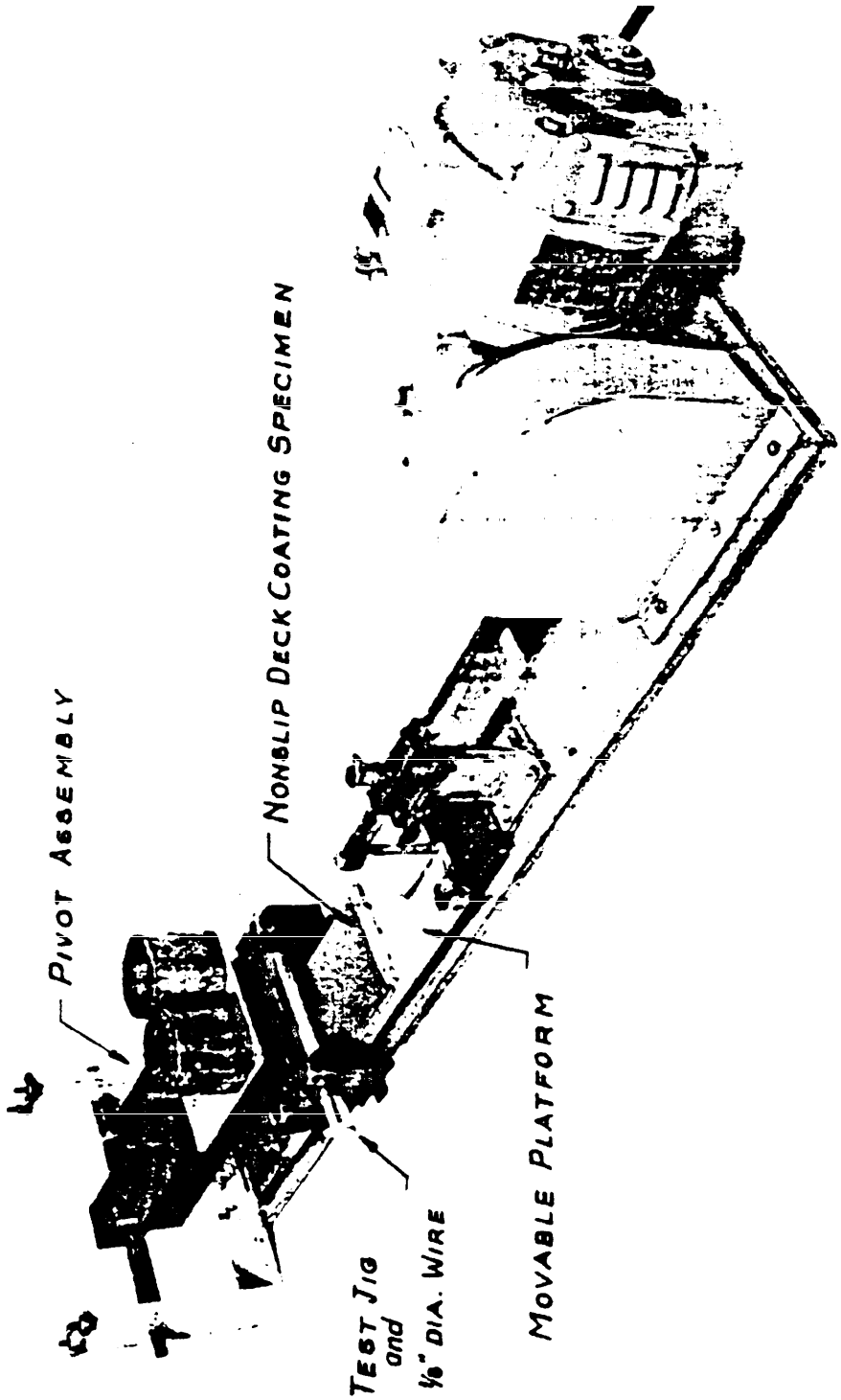
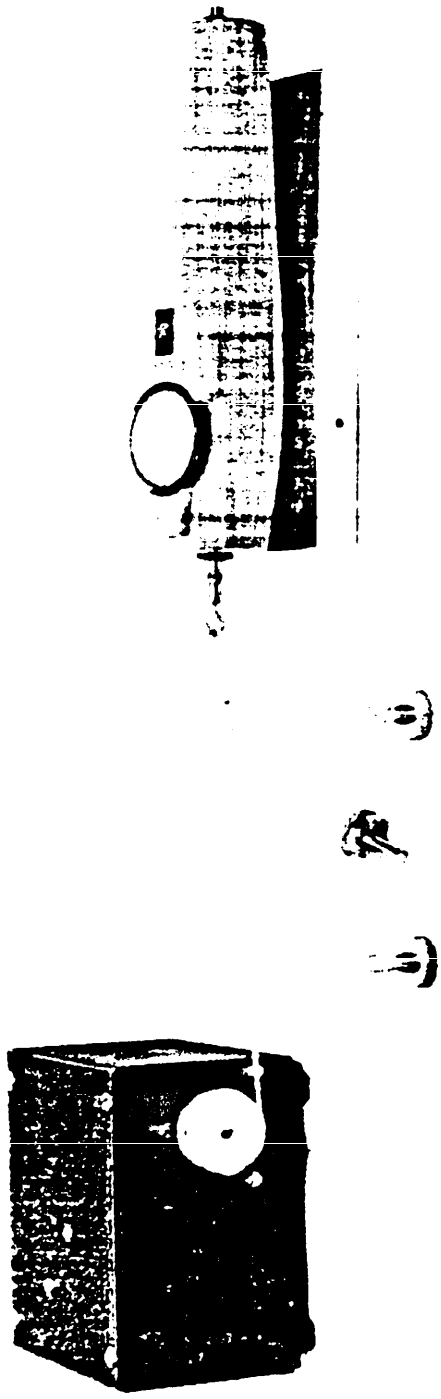


FIGURE 1. Arresting gear cable wear test apparatus.

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SH 9017

FIGURE 2. Coefficient of static friction apparatus (Olson Slipmeter).

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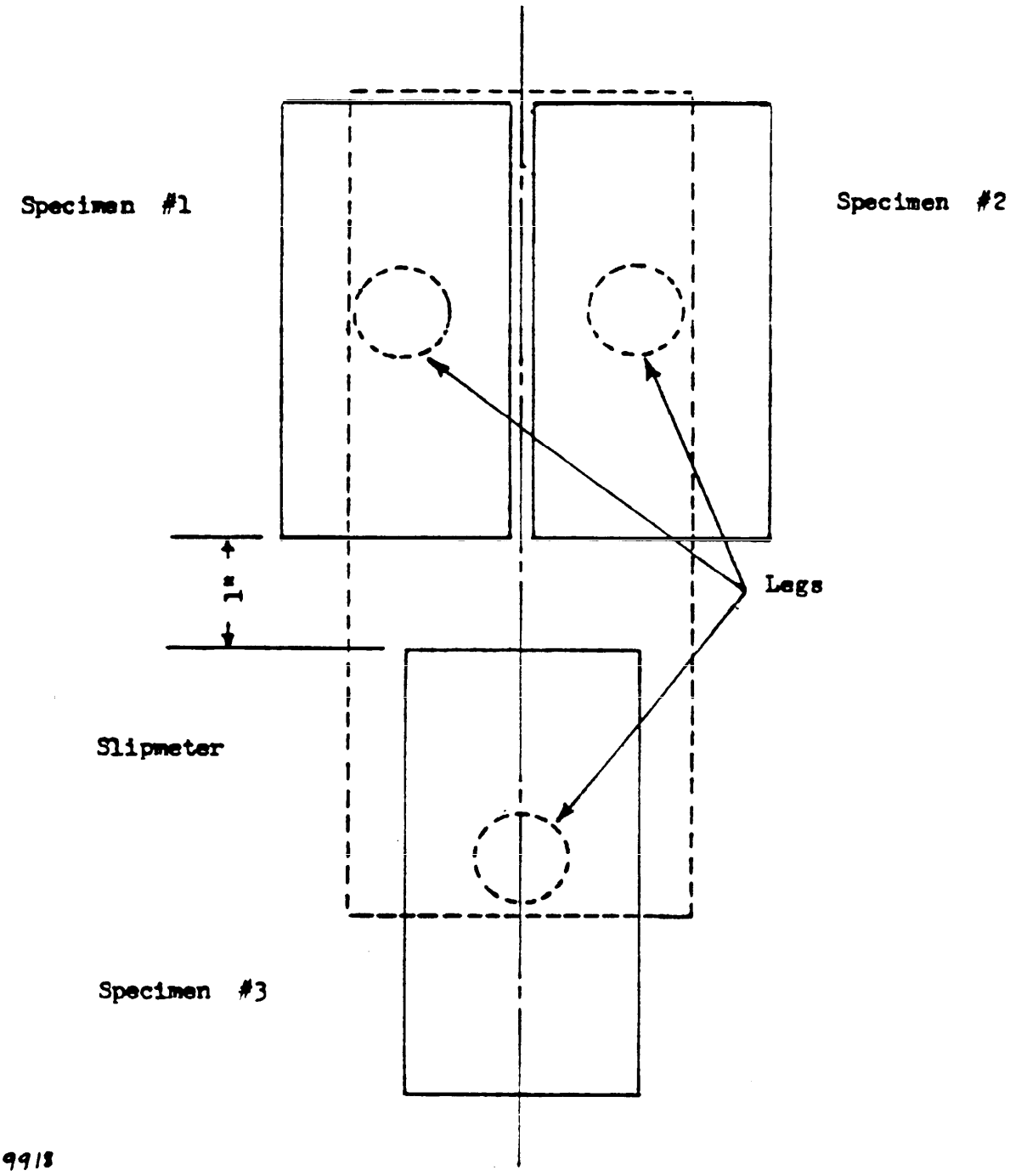


FIGURE 3. Arrangement of 2 by 3 inch specimens for measurement of coefficient of friction.

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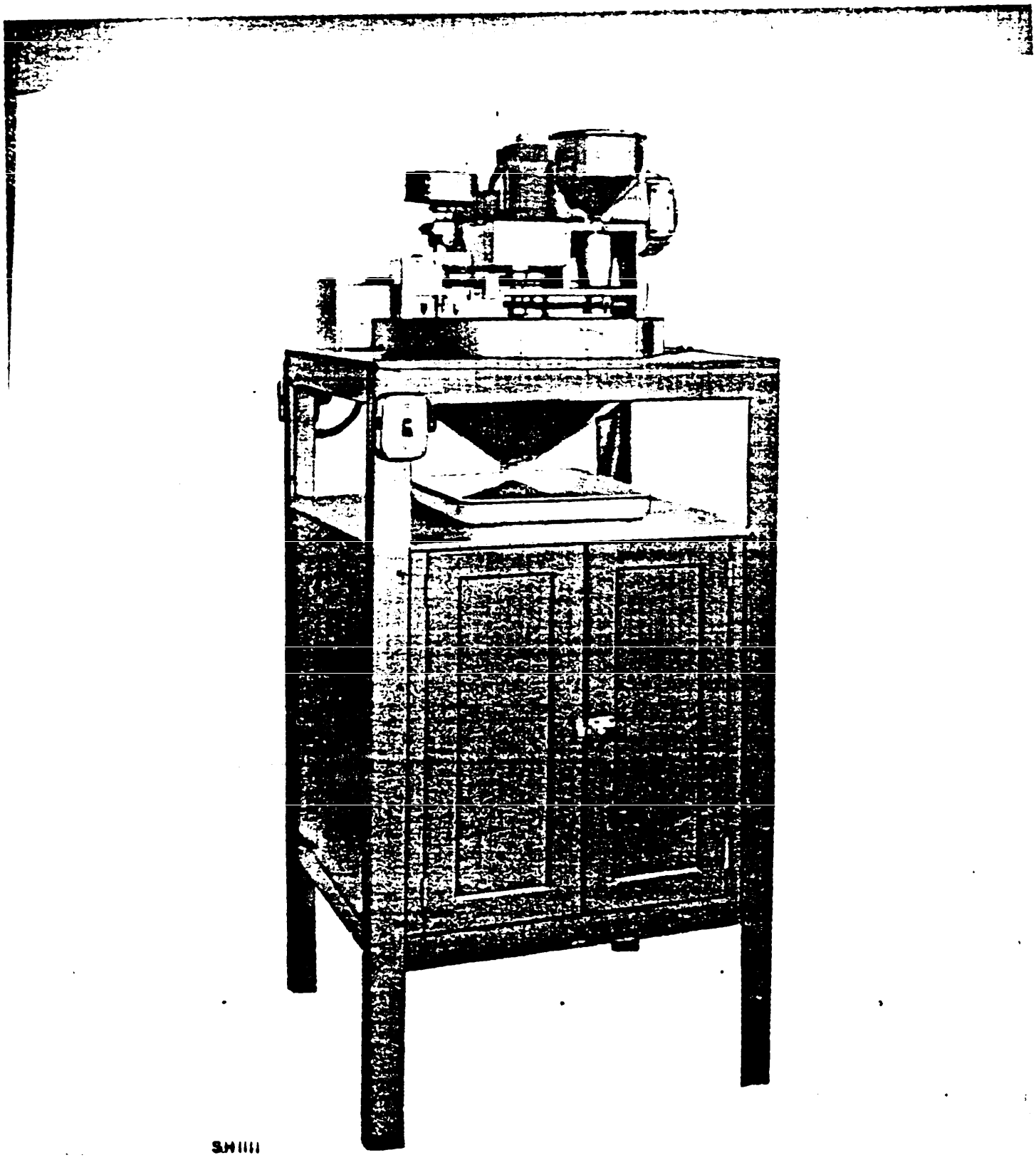


FIGURE 4. Wear test machine.

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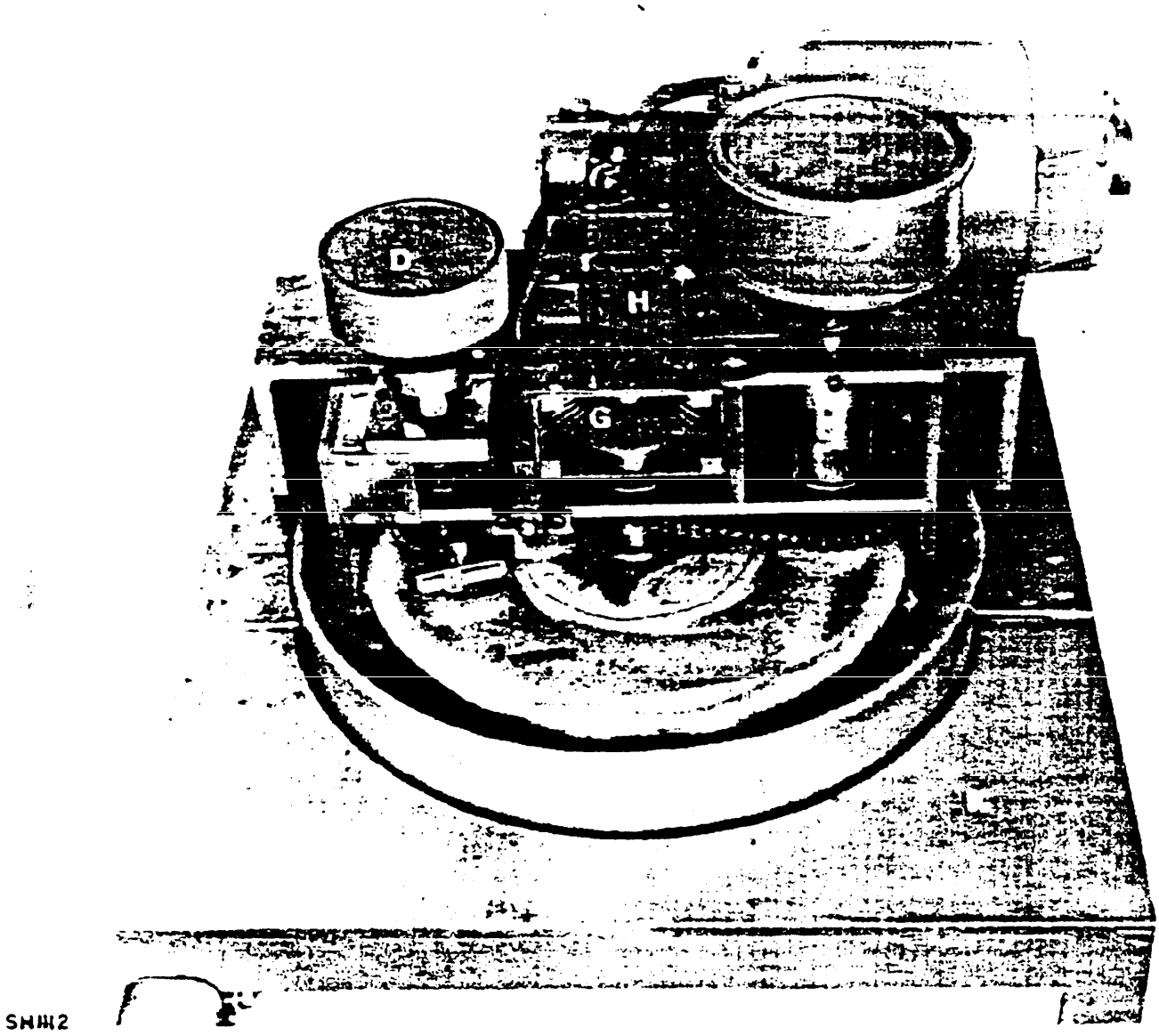


FIGURE 5. Detail view of wear-test machine.

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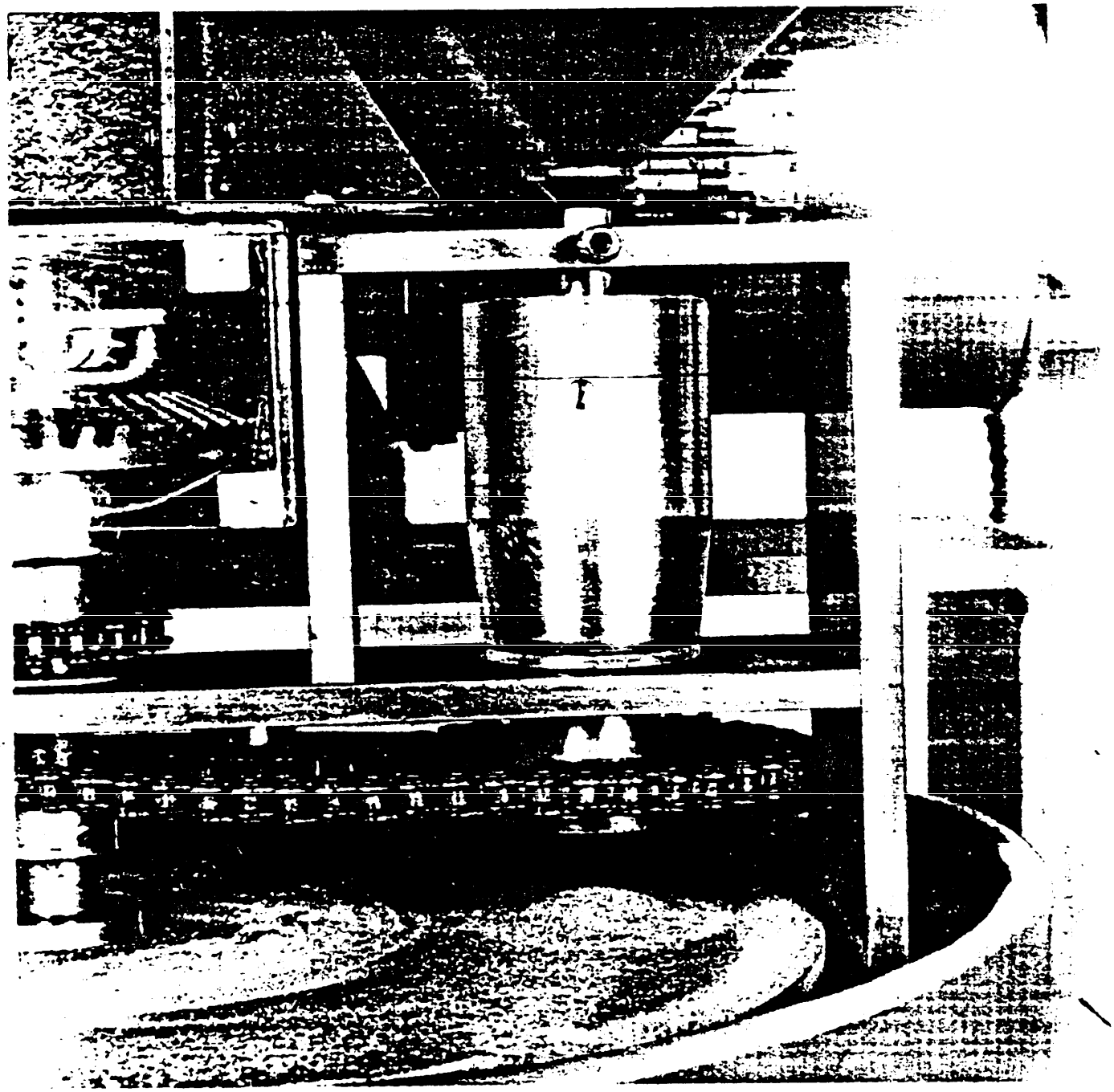


FIGURE 6. Detail of distributor of abrasive grit.

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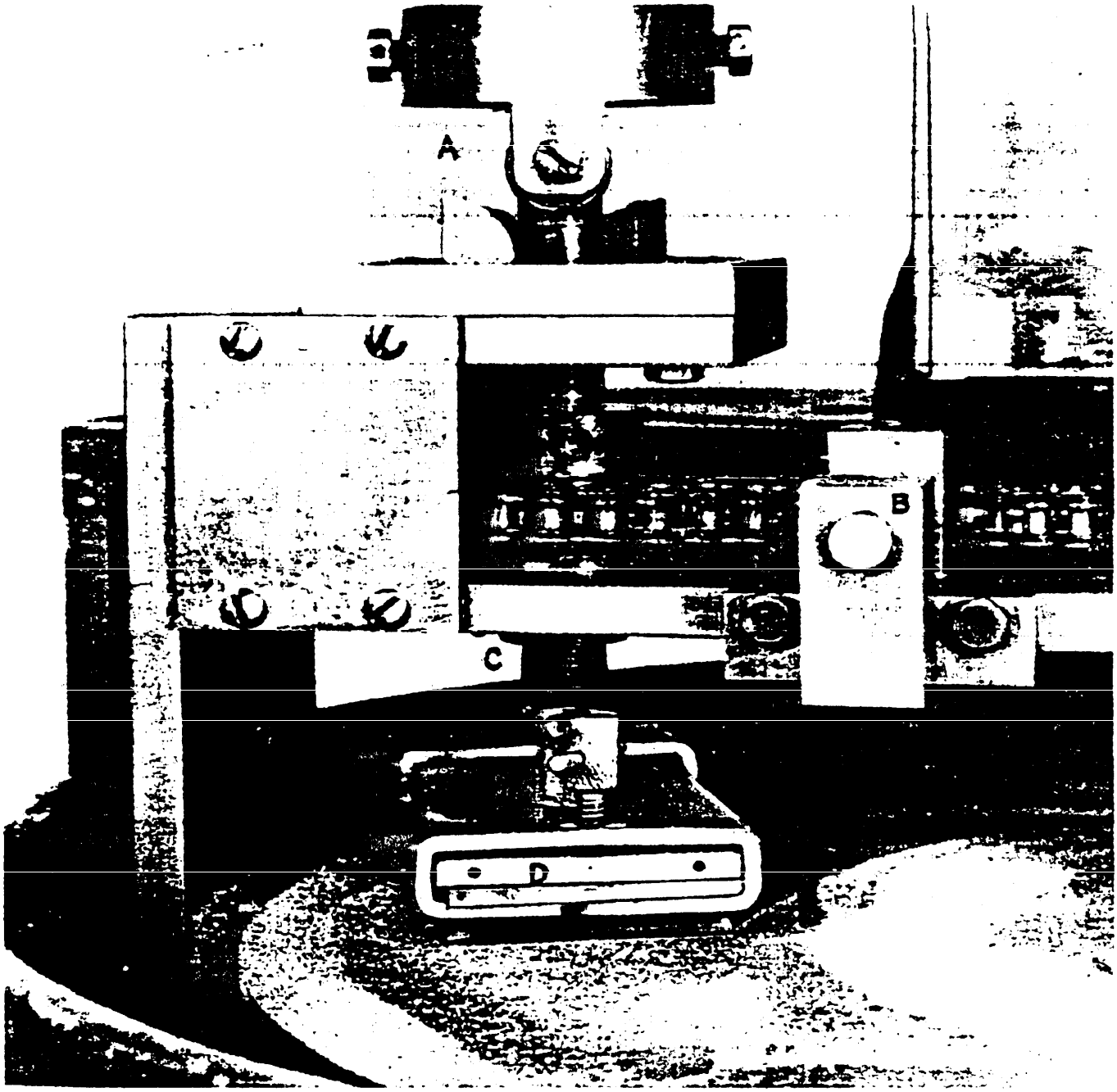


FIGURE 7. Detail of cam and specimen holder.

A - Cam
B - Chain idler
C - Universal joint
D - Specimen holder

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